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ABSTRACT: Like nearly all biologic structures, the peripheral nervous system is remarkably temperature sensitive. Clinical neurophysiologists are most aware of the untoward effects of cooling on nerve conduction studies, including reduced conduction velocity, prolonged distal latency, and increased response amplitude and duration. However, familiarity with the effects of temperature variation on the peripheral nervous system can also provide a deeper understanding of the physiological mechanisms underlying the function of nerve, muscle, and neuromuscular junction in health and disease. Intentional temperature alteration can also improve the diagnostic accuracy of certain electrophysiologic tests, such as the use of heat when performing repetitive nerve stimulation in myasthenia gravis or the use of cold during needle electromyography in some of the myotonic disorders. Finally, extremes of temperature have long been known to produce permanent neuronal dysfunction; recent investigations are beginning to elucidate the mechanisms of such injury.

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EFFECTS OF TEMPERATURE ON NEUROMUSCULAR ELECTROPHYSIOLOGY

SEWARD B. RUTKOVE, MD

Department of Neurology, Beth Israel Deaconess Medical Center, Harvard Medical School, 330 Brookline Avenue, TCC-810, Boston, Massachusetts 02215, USA

Temperature variation has profound effects on the peripheral nervous system. This is not surprising given the inherent temperature sensitivity of many of its underlying structures, such as voltage-gated ion channels, acetylcholinesterase, and the contractile apparatus of muscle. As each component demonstrates a unique profile of temperature sensitivity and as any temperature change will inevitably affect multiple components simultaneously, the overall effects of cooling or heating can be quite complex.

Most clinical neurophysiologists view temperature fluctuations as a nuisance, and for good reason. As most normal values have been determined within a relatively narrow temperature range (32–36°C), daily experience with temperature is focussed on keeping a limb warm from the beginning to the end of electrophysiologic testing. For example, most consultants have seen the exceptionally anxious patient who presents with cool, clammy hands on a winter day. Despite aggressively trying to warm the limbs with hot packs or a hot-water bath, the hands quickly cool during the study. In the end, the results of the testing may be significantly compromised.

The primary goal of this minimonograph is to review these temperature-related alterations in electrophysiologic parameters. A secondary aim is to demonstrate how an understanding of the effects of temperature fluctuation can provide valuable insights into the function of the peripheral nervous system and muscle in both health and disease. This article begins with a review of the practical aspects of temperature measurement and manipulation followed by a discussion of temperature effects on the peripheral nervous system in health and disease. The article concludes with a review of mechanisms underlying the nerve injury that occurs with exposure to extreme heat or cold.

Abbreviations: ACh, acetylcholine; ATPase, adenosine triphosphatase; CMAP, compound muscle action potential; CV, conduction velocity; EDX, electrodiagnostic medicine; EMG, electromyography; EPP, endplate potential; *m,* number of ACh quanta; MEPP, miniature endplate potential; MUAP, motor unit action potential; *n,* magnitude of the store of readily available quanta; *p,* probability of release; SRMAP, spontaneous repetitive motor action potential

Key words: cold; electrophysiology; heat; muscle; nerve; neuromuscular junction; neurophysiology; temperature

Correspondence to: American Association of Electrodiagnostic Medicine, 421 First Avenue SW, Suite 300 East, Rochester, Minnesota 55902; e-mail: aaem@aaem.net

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PRACTICAL ASPECTS OF TEMPERATURE MEASUREMENT AND MANIPULATION

Temperature Measurement. The accurate measurement of nerve or muscle temperature represents

a challenge in itself. Generally, clinical neurophysiologists rely on surface measurement to give an approximation of the true temperature of a nerve or muscle. All in all, this works reasonably well. In one study using a thermostatically controlled heat source, it was found that a surface temperature of 37–38°C corresponded to a near-nerve temperature of 36–37°C.[6] In another study, near-nerve temperature was found to differ from surface temperature by up to 1.7°C.[27] At colder temperatures, the surface reading was usually lower than the near-nerve temperature, whereas the reverse was true at warmer temperatures. Given the variety of shapes and sizes of individual limbs, it is reasonable to expect that the correlation between surface and near-nerve temperature will vary to some extent from person to person. Of course, even a near-nerve temperature obtained with a needle thermometer has its limitations, as only one segment of nerve is being measured, and it is likely that some variation in temperature will be present along the nerve.

External heating or cooling of a limb can also represent a potential source of error, as the surface temperature will change rapidly, whereas the near-nerve temperature will adjust more slowly.[56] Thus, whether warming or cooling, sufficient time is necessary to allow the internal temperature to "catch up" with the external temperature. Although some authors have suggested that 30 min is preferable,[48] this length of time is generally impractical, and 10 to 15 min is usually sufficient.

Whether internal or external, temperature is usually measured with either a thermistor or thermocouple, both of which utilize changes in electrical resistance to determine temperature. It is worth noting that the size of the thermistor or thermocouple used will also vary directly with the length of time it takes for temperature readings to stabilize, with a small device having the most rapid response time.

"Ideal" Temperature. Electrodiagnostic medicine (EDX) consultants generally try to maintain the upper extremities between 34°C and 36°C and the lower between 32°C and 34°C. However, these should not be viewed as simply ideal temperature ranges at which to perform nerve conduction studies. Rather, they represent a convention of sorts, so that one can interpret results from any electromyography (EMG) laboratory or compare results on the same patient from two different examination dates. In point of fact, higher temperatures could theoretically improve the sensitivity of certain tests, such as 3-Hz repetitive stimulation in myasthenia gravis or the detection of conduction block in a patient with a demyelinating polyneuropathy, as discussed in the section on Temperature Effects in Neuromuscular Disease.

Heating/Cooling Methodologies. One of the major difficulties in studying the effects of temperature on nerves is the inherent problem in adjusting the temperature of a limb. Given variations in girth, initial temperature, the exact location of a given nerve or muscle, and the patient's vasodilatory state, the same heating or cooling technique used on two individuals will almost always produce slightly different results.

Patients may present to the EMG laboratory with excessively cool extremities for many reasons, the most obvious of which is a cool environment. However, patients may also have reduced body fat and muscle, making maintenance of temperature more difficult. Finally, limbs may cool in the EMG laboratory secondary to anxiety-induced sympathetic activation.

Perhaps the most effective warming technique is that of immersing the limb to be studied in a hot-water bath. Chemical hot packs, such as those using calcium chloride, and reusable microwave packs are also effective, though they do not provide as rapid, homogeneous heating as does a water bath. Electrical heating pads are usually avoided because of the prominent electrical interference they create, although they can be helpful in warming the limb prior to study. An infrared heating apparatus can also work well, especially with an attached thermocouple placed on the forearm to better control the temperature. Hair dryers are used, as well, in some laboratories. One final methodology for heating that has been only minimally explored is therapeutic ultrasound. Studies have demonstrated that this modality can increase deep-tissue temperature quite effectively.[37,59,81] Regardless of the exact methodology used, care must be exercised to avoid burns in patients with pre-existing nerve injury in whom extreme temperature cannot be appreciated. Also, monitoring the temperature during a study is important, as extremities may cool rapidly.

The application of ice or cold water remains the usual means for cooling an extremity when performing research investigations. An ice pack is placed intermittently on the area to be studied while temperature is continuously monitored. Another method of cooling (or heating, for that matter) most easily applied in the research setting is the use of a cooling electric block or coil. Like heating, excessive cooling can produce potentially permanent injury (see section on Neuronal Dysfuction and Injury Due to Tem-

perature Extremes, below) and it is important to monitor temperature and time exposure closely.

Focal versus Diffuse (Segmental) Cooling/Heating. Exactly how a limb is cooled or heated also will have dramatically different effects on neurophysiologic measurements. In focal cooling, for example, a small area only is cooled, often at the recording site. An example would be placing ice directly on digit 2 while performing an antidromic median sensory nerve conduction study recording from that same finger. In this situation, the digital nerves cool profoundly, but the proximal median nerve remains at standard temperature. In diffuse or segmental cooling, rather than just isolating the cooling to the finger, the entire limb is cooled equally. For example, the entire forearm, hand, and finger would be cooled in an ice bath and would be at identical temperature.

Clearly, neither form of cooling is exclusive. When a patient presents to the EMG laboratory after being outside on a frigid winter day, usually the most distal aspects of the hands and feet are the coldest, with a gradient of increased warmth proximally. This graded temperature change actually represents a mixture of focal and diffuse cooling. Even in the research setting, it is difficult to change the temperature of one region of nerve or muscle and leave an adjacent area unchanged. The importance of distinguishing between these two types of temperature variation will be explained in detail below.

Correction Factors. The use of simple extrapolation to correct for temperature variance is an appealing concept. If one can predict the degree of conduction velocity (CV) slowing from a reduced temperature, the EDX consultant would not need to warm the limb and would thus save time and effort.

Initial reports of relatively linear reductions in CV with decreasing near-nerve temperature suggested that this might be feasible. Henriksen[63] described a reduction of 2.4 m/s per degree Celsius between 38°C and 29°C for the median and ulnar nerves. Later, Gasser and Trojaborg[53] found a lower correction factor of 1.7–1.9 m/s per degree Celsius for the near-nerve temperature of the sciatic nerve.

Other authors[1,57,58] have attempted to determine correction factors for the more easily obtained surface temperature. Most have found a 1.5–2 m/s reduction in CV per 1°C variation in surface temperature. For example, Halar and coworkers[57,58] found 2.1 m/s per degree Celsius change in ulnar motor and 1.6 m/s per degree Celsius in ulnar sensory studies and a 2.0 m/s per degree Celsius change in the peroneal motor studies. Likewise, Edelwejn[40] found a 1.5 m/s per degree Celsius reduction in CV of the median nerve. This temperature sensitivity likely varies between fiber class, with larger-diameter, faster-conducting fibers having higher temperature sensitivity.[36,52,54] In order to circumvent this issue, De Jesus and colleagues[27] suggested a semilogarithmic relationship utilizing the Q_{10}, the ratio of the highest to lowest velocity over a 10°C range.

Despite all this study, however, correction factors have significant limitations. The first is that interpatient heterogeneity is always present and a correction factor for one individual may not hold for another. Second, such factors do not take into account changes in amplitude or area. Finally, and most importantly, these correction factors are reasonably accurate only in health. Predictions in diseased states may not hold. In fact, as described in a later section, there is evidence that the CV in demyelinated nerves is likely to exhibit reduced temperature sensitivity as compared with normal nerves.

In short, the use of correction factors to adjust CV for temperature variation is acceptable only in the occasional circumstance that limb temperature cannot be normalized. In all other situations, warming of the limbs into the standard range will allow the electrophysiologist to interpret nerve conduction study results with the most confidence.

TEMPERATURE EFFECTS IN HEALTH

Table 1 summarized the effects of cooling on the neuromuscular system.

Nerve.

Resting Membrane Potential. The permeability of the cell membrane to various ions and the concentrations of those ions inside and outside the cell produce a net potential difference across the membrane. K^+ is sequestered within the cell while Na^+ is pumped out, resulting in a resting membrane potential of approximately −60 mV for neurons.

Temperature appears to have only a modest effect on neuronal resting membrane potential. In the squid axon, the resting membrane potential is remarkably stable between 3°C and 20°C. By 35°C, it diminishes (becomes more positive) by only 10–15 mV.[66] In two studies looking at rats, one showed a stable muscle membrane potential down to 10°C,[90] whereas another reported a direct linear relationship with a small Q_{10} of 1.2.[101]

Action Potential in Single Neurons. As Hodgkin and Huxley first demonstrated in 1945, the depolarization of a single nerve axon is due to the abrupt

Table 1. Cooling effects in health.

Basic parameters	
Resting membrane potential	Mild ↓
Na^+/K^+ pump activity	↓
Rate of neuronal depolarization	↓
Total inward Na^+ flux	↑
Total outward K^+ flux	↑
Refractory period	↑
Ca^{++} half-life in nerve terminal	↑
MEPP amplitude	↑
ACh release with single stimulus	↓
Nerve Conduction Studies	
Sensory amplitude	Focal, ↑; Diffuse, ↑/NC
Motor amplitude	Focal, Mild, ↑; Diffuse, NC
Duration	Focal, ↑; Diffuse, ↑
Conduction velocity	Focal, NC; Diffuse, ↓
Needle EMG	
MUAP amplitude	Conflicting results
MUAP duration	↑
MUAP phases	↑
Additional Tests	
3-Hz repetitive stimulation	NC with mild cooling
Jitter	Not established
CMAP amplitude increase after exercise	↓
Muscle Contraction	
Duration contraction/relaxation	↑
Force	↓

NC, no change.

inward flux of Na^+ ions. This influx of Na^+ ions may occur due to the opening of voltage-gated channels along the axon or to the binding of a neurotransmitter (as occurs on the muscle endplate). Following the initial influx of Na^+, the Na^+ channel closes and K^+ channel opens, allowing for a small efflux of potassium, restoring the membrane to its resting potential. K^+ efflux plays only a minor role in mammalian axon's repolarization as compared with those of invertebrates, where the initial work by Hodgkin and Huxley was performed. Repeated depolarizations of a cell will eventually decrease the potential gradient. In order to counteract this decrease, the Na^+/K^+ adenosine triphosphatase (ATPase) is constantly pumping out Na^+ and bringing in K^+, helping to maintain the resting potential.

It is the configuration of the action potential that is most affected by temperature variation. A decrease in temperature slows down both the opening and closing of the Na^+ channel, with the closing being more affected than the opening, as Hodgkin and Katz[66] first demonstrated in 1949. This phenomenon has several effects. First, the slower depolarization translates into a reduced CV for the axon studied. Second, the slower depolarization and repolarization produces a longer channel-open time and, therefore, a longer duration response. Finally, as additional Na^+ ions move into the cell during the prolonged opening, a larger depolarization occurs and the response amplitude increases. Heating, of course, has the opposite effect. With temperature elevations, Stegeman and De Weerd, [123] using computer modeling, demonstrated increased CV and decreased action potential duration in models of single sensory fibers; amplitude measurements, however, were less consistently altered.

The absolute and relative refractory periods also will lengthen with cooling.[29,92] For example, Lowitzsch and colleagues[93] found that the absolute refractory period increased from 0.54 ms at 35°C to 3.07 ms at 20°C, and the relative refractory period increased from 3.19 ms to 20.09 ms over the same temperature range. The likely explanation for these effects is that, like the open time, the reactivation time is also increased with cooling, preventing repeated depolarization for a prolonged period.

Sensory Conduction Studies. When performing standard sensory nerve conduction studies, EDX consultants do not study just a single axon. Rather, populations of axons are evaluated, because temporal dispersion and phase cancellation play important roles. Hence, the changes described above become more complex and are dependent, in part, on the specific type of cooling or heating (focal versus diffuse) applied to the limb.[82]

An example of an antidromic median sensory nerve conduction study stimulating at the wrist and recording from digit 2 can help clarify these temperature effects (Fig. 1). Focal cooling of digit 2 will have a relatively straightforward effect. As only the nerve fibers directly under the recording electrodes will be cooled, the CV in the forearm and hand will remain unchanged. However, when the depolariza-

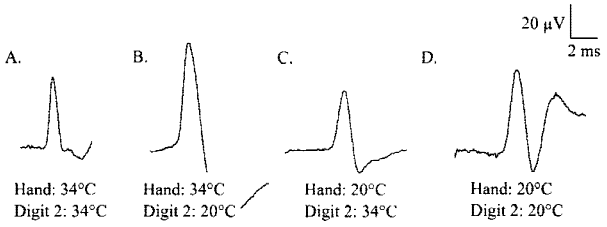


FIGURE 1. Effects of different types of cooling on the median compound sensory nerve action potential recording at the wrist and recording from digit 2. **(A)** Baseline study at 34°C. **(B)** Focal cooling restricted to digit 2 demonstrating a predominant effect on response amplitude and, to a lesser extent, duration. **(C)** Cooling restricted to wrist and hand with digit 2 maintained at 34°C, demonstrating prolongation in distal latency and mild increased response duration with minimal change in amplitude from baseline. **(D)** Diffuse cooling of wrist, hand, and finger, demonstrating a combination of the changes demonstrated in **B** and **C**. See text for full explanation.

tion reaches the finger, it will abruptly slow under the recording electrodes, producing a response with an increased amplitude and duration (Figs. 1A and 1B).

With diffuse cooling, however, the changes are more complex. A variation of diffuse cooling, in which digit 2 is maintained at 34°C, is considered first. As the nerve is cooled, the CV of all the neurons will begin to slow. However, this will cause the resulting compound sensory potential to disperse, as the slower fibers will be more affected than will the faster. For example, two axons, one conducting at 60 m/s and another at 40 m/s at 35°C are studied. Over a distance of 30 cm, the fast action potential will take 5.01 ms, whereas the slower will take 7.50 ms, leaving them 2.49 ms apart. If they both slow by 1.5 m/s per degree Celsius with cooling, at 25°C the fastest will now be traveling at 45 m/s, taking 6.66 ms, and the slower will go at 25 m/s, taking 12.0 ms. The two potentials will now be 5.34 ms apart, almost 3 ms further apart than at 34°C. This enhanced dispersion will cause the compound response duration to increase and amplitude to decrease while the overall CV slows (Fig. 1C).

When cooling of digit 2 occurs simultaneously with that of the hand, as is seen in true diffuse cooling, a combination of effects is observed, with only a modest increase in response amplitude and duration as compared with baseline and a slowed CV (Fig. 1D).

Not surprisingly, these differences have led to inconsistent results between studies. For example, Ludin and Beyeler[94] found a relatively linear effect on orthodromic sensory CV and duration, with gradual warming from 22°C to 36°C, but that compound sensory amplitude appeared to reach an apex at 26°C. On the other hand, Bolton and colleagues[11] found a relatively linear affect for all parameters, including amplitude. The authors also confirmed an earlier observation by Hlavova and colleagues[65] that orthodromic sensory responses are much less temperature sensitive than are antidromic responses, possibly due to insulating effects at the recording site in the former method. Dioszeghy and Stålberg, [34] applying diffuse cooling, found a trend toward larger response amplitudes in normal subjects, but the change did not reach statistical significance. Rutkove and colleagues[113] found a 50% reduction in antidromic sensory response amplitude with diffuse heating of the arm (including the recording site) from 34°C to 42°C.

Muscle. ***Resting Membrane Potential and Action Potentials.*** Muscle fibers maintain a resting potential which is hyperpolarized relative to nerve, at approximately –90 mV. Like neurons, temperature variation produces only a mild effect on the resting membrane potential of muscle fibers at rest. In one study of rat myotubes, a resting potential of –90mV was present at 35–37°C; with cooling to 10–15°C, a partial depolarization ensued, with the membrane potential reaching –60 mV.[14] This change is due likely to decreased activity of the Na^+ K+ ATPase with cooling.[35]

Like neurons, the depolarization of muscle fibers depends on the influx of Na^+ ions. Therefore, it is not surprising that cooler temperatures prolong the depolarization of individual muscle fiber potentials and slow the muscle fiber CV. However, as EDX consultants generally examine motor unit action potentials (MUAPs) and compound motor action potentials (CMAPs) rather than individual muscle fiber action potentials, these temperature effects will be observed as increased duration and amplitude of CMAPs and of individual motor units on needle EMG (see below).

Motor Conduction Studies. The changes that occur in motor conduction studies with temperature fluctuation are actually quite similar to those that occur in sensory studies. Once, again, the specific character of the heating or cooling applied is important, with focal distal cooling having a more profound effect on amplitude and diffuse cooling reducing CV. Similarly, diffuse cooling will also prolong the distal latency, although some slight prolongation may also be evident with focal distal cooling due to some inevitable spread of the reduced temperature proximally. Overall, these temperature-related effects tend to be more modest than those observed in sensory studies. Bolton and colleagues[11] demonstrated a reduction in response amplitude with increasing temperature. However, Dioszeghy and Stålberg[34] could not find a significant change. Both studies, however, demonstrated reductions in motor CV and increased CMAP duration with cooling. With heating, the opposite effects occur. With diffuse heating of the arm to 42°C, for example, Rutkove and colleagues[113] identified an average reduction in ulnar motor amplitude of 27%, a 19% reduction in CMAP duration, and an 11% increase in CV.

Electromyography. *Insertional Activity.* As the resting membrane potential does not change markedly with temperature variation, there is no significant change in character of insertional activity identified in normal muscle.

Motor Unit Action Potentials. Focal cooling of a muscle will affect the configuration of a MUAP. A motor unit is made up of the anterior horn cell, its

axon with its terminal branches, and the individual muscle fibers attached to those branches. The potential recorded represents the depolarization of those muscle fibers closest to the recording electrode. Cooling of muscle might be anticipated to produce MUAP changes parallel to those observed in the CMAP. Specifically, focal cooling should slow the CV of action potentials in terminal nerve twigs (increasing desynchronization and hence polyphasia) and lengthen the duration and increase the amplitude of depolarizing muscle fibers. Indeed, Buchthal and colleagues[17] found an increase in the mean duration of biceps MUAPs with cooling. The number of polyphasic potentials also increased, with 3% at 37°C to 10–15% at 30°C and 30–45% at 24–25°C. However, in their study, motor unit amplitudes actually decreased 2–5% per degree Celsius relative to 37°C. In another study of single motor units at different temperatures, Falck and Lang[44] reported a 4% increase in total and spike duration at temperatures below 35°C in tibialis anterior, with amplitude increasing 0.6% per degree Celsius. More recently, Bertram and colleagues[9] used automatic decomposition EMG to study this question, as well, and they found an inverse relationship between MUAP duration and temperature, with no significant change in turns or MUAP amplitude with focal muscle cooling.

Muscle Contraction. Muscle contraction occurs when the depolarization of the muscle membrane leads to the release of Ca^{++} from the sarcoplasmic reticulum, producing binding between the actin and myosin filaments and increased tension within the muscle. Ca^{++} is then taken back up into the sarcoplasmic reticulum and the muscle contraction ends. Given that most biochemical processes slow with cooling, the a priori expectation would be that of a slower, more prolonged muscle contraction–relaxation cycle with cooling. Indeed, one early study demonstrated that contraction and relaxation were both slowed equally[64]; however, in another study, cooling appeared to slow muscle relaxation more than contraction.[125]

Studies have demonstrated that cooling reduces the maximal force generated by a muscle contraction, with maximal peak strength during dynamic contractions falling by 2–5% per degree Celsius[8,23,116] Endurance time, however, has been found to be longer at cooler temperatures than at warmer.[68] The mechanism for these changes is unclear. Desynchronization in the excitation of individual muscle fibers is one possible factor, but a direct effect on the contractile mechanism or muscle metabolism is also likely.

After-Effects of Contraction. After brief exercise, the CMAP often increases in amplitude, an effect termed "pseudofacilitation." The mechanism of this phenomenon has been debated, with changes in the synchrony of motor unit depolarization and in Na^+/K^+ ATPase activity proposed as possible explanations. If a brief isometric contraction is performed by a muscle that has been cooled, this effect is no longer apparent. Rather, the obtained response will be of lower amplitude than the initial one and of markedly longer duration.[111] Heating produces only subtle changes. Although the mechanism of this temperature-related effect is uncertain, changes in Na^+/K^+ ATPase activity are most likely playing a role in mediating these effects.

Neuromuscular Junction. The neuromuscular junction is the most physiologically complex structure routinely studied in the EMG laboratory. The multiple processes that occur there, including influx of Ca^{++} into the nerve terminal, binding of vesicles for release of acetylcholine (ACh), binding of ACh on the postsynaptic membrane, the formation of the endplate potential (EPP), and hydrolysis of ACh by acetylcholinesterase all demonstrate some degree of temperature sensitivity. The cumulative result of these different effects is that cool temperatures tend to enhance neuromuscular junction transmission.

Spontaneous Release of ACh Quanta. Increasing temperature enhances the release, and thus the frequency, of miniature endplate potentials (MEPPs) in mammalian muscle.[13,45,91] When two molecules of ACh bind to the endplate receptor, an influx of Na^+ occurs. With cooling, the duration of the channel open time increases, similar to that which occurs in nerve and muscle membrane, resulting in an increase in the amplitude of an individual ACh-induced depolarization.[85] Likewise, as the MEPP consists of many ACh-induced depolarizations, cooling increases its amplitude, as well. However, the increase is less than might be anticipated based on changes in depolarization of individual channels alone. Although the explanation for this inconsistency is uncertain, it may be due to a decreased number of ACh molecules binding to the postsynaptic membrane per quantum released.

Stimulus Evoked Responses. Endplate potentials result when a number (m) of ACh quanta are released and bind to the muscle endplate after a neuronal depolarization. When the EPP is of sufficient magnitude, it exceeds the threshold for muscle depolarization and causes a propagated muscle action potential. The release of ACh is a function of the probability of release (p) and the magnitude of the

store of readily available quanta (n); both increase with elevated temperature.[67,71] For example, the number of ACh quanta released into the rat neuromuscular junction in one study was 107 at 26°C and 165 at 37°C.[67] However, the complexity of this analysis increases when stimuli are delivered at different frequencies. This is true because facilitation (the increase in ACh release due to increased levels of presynaptic Ca^{++}) and replenishment of stores (that is, movement of ACh from secondary to primary stores) come into play. In healthy nerves, increasing temperature has opposite effects: hastening the replenishment of transmitter in the presynaptic terminal but decreasing the duration of facilitation.[3,13,28,67,70,71] The latter effect is due to a reduced half-life of Ca^{++} in the nerve temperature with heat.

However, elevations in temperature, at least to a point, also increase the activity of acetylcholinesterase,[47,119] decreasing the number of ACh molecules available for binding to the endplate. Increasing temperature also will reduce the amplitude and duration of a given ACh-induced depolarization,[60,85] once again due to reduced Na^{+} influx, making it more difficult to produce a muscle fiber action potential.

Ultimately, the multiplicity of effects of temperature on the neuromuscular transmission leaves any analysis oversimplified. The bottom line, however, is that in normal subjects, these temperature-dependent changes have no effect on the outcome of neuromuscular junction testing. Repetitive stimulation at slow rates will remain normal with moderate cooling and heating.[113] In part, this is due to a very high safety factor, with far more ACh released with a given stimulus than is necessary to induce a muscle fiber depolarization.

Jitter. Jitter in normal subjects might be expected to decrease with higher temperature and increase with cooler, as at higher temperature ACh release would likely be more synchronous between neuronal depolarizations. Unfortunately, data to validate this hypothesis have not been published. It is worthwhile to note, however, that in the published normal values, the highest limits on normal jitter are in distal muscles.[15] Although the reason for this is debatable, it may be due to variations in temperature, with distal muscles being slightly cooler than proximal, as temperature standardization across muscles was not specified.

TEMPERATURE EFFECTS IN NEUROMUSCULAR DISEASE

Table 2 summarizes the effects of temperature changes in disease.

Neurogenic Disease.

Axonal Loss Disorders. *Nerve Conduction Studies.* It is clear that focal cooling will increase the amplitude of both motor and sensory responses in patients with axonal loss lesions, just as occurs in normal subjects. However, in certain situations, this temperature sensitivity appears to be heightened. For example, Denys[32,33] demonstrated that in patients with early motor neuron disease, cooling produced an unexpected increase in the response area, much greater than that seen in normal subjects (Fig. 2). This was especially true in patients with more mild disease. One explanation for this temperature sensitivity with focal cooling is that when fewer motor units are present, temporal dispersion and phase cancellation are less likely to play as significant a role as when there are numerous motor units present. Improved neuromuscular transmission with cooling could also play a role. This response to focal cooling has not been pursued in other disorders characterized by sensory or motor axonal loss.

Diffuse cooling also may have unique effects in patients with axonal loss diseases, but it has not been studied extensively. Although Dioszeghy and Stålberg[34] studied individuals with axonal loss lesions, the small number and heterogeneity of patients make it difficult to draw any firm conclusions.

Electromyography. Motor Unit Action Potentials. Although no study has addressed this question

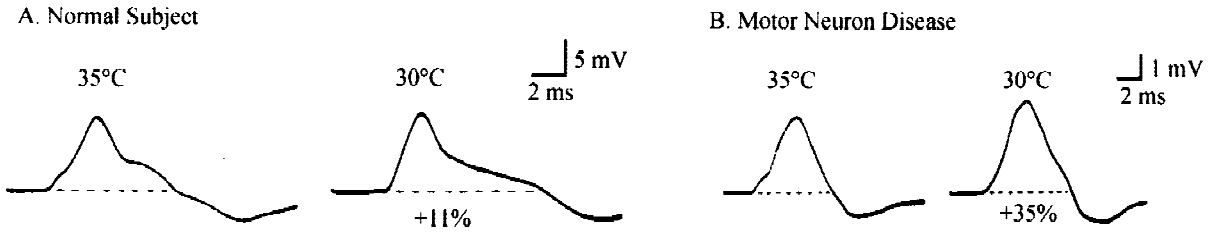


FIGURE 2. Effect of focal cooling on the CMAP in a normal subject **(A)** and in a patient with motor neuron disease **(B)**. Note how focal cooling increases CMAP amplitude relatively more in the patient than in the normal subject. Modified from Denys, 1991.[31]

specifically, as cooling prolongs the MUAP duration in normal muscle, it probably has the same effect in reinnervated muscle. Heating would likely have the opposite effect.

Fibrillation Potentials and Positive Sharp Waves. Fibrillation potentials and positive sharp waves represent the spontaneous, repetitive depolarization of single muscle fibers. Although some controversy persists regarding their different morphologies, most agree that a positive sharp wave occurs when the EMG needle is in close proximity to the muscle fiber being recorded, with the depolarization aborting in the region of muscle fiber near the needle.[38] Regardless of the specific differences between these two wave forms, both occur in partially depolarized muscle fibers.

Langley[83] first studied the effect of cooling on muscle in cats. By visual inspection alone, the author observed that fibrillation potentials increased in frequency with warmth and decreased with cold. Feinstein and colleagues[46] studied local as well as generalized temperature changes using both simple visual observation of the muscle as well as needle electrodes, confirming the findings of Langley. Recently, Lee and Kwon[86] were able to semiquantify the frequency of fibrillation potentials with decreasing temperature in rats, demonstrating their disappearance in temperatures below 20°C.

The importance of this effect in the clinical arena is obvious, as the diagnosis of many neuromuscular diseases is based on the objective finding of fibrillation potentials. Denys,[31] in an earlier version of this minimonograph, underscored this by describing a case of an elderly gentleman with generalized weakness in whom initial needle EMG was relatively unremarkable, as the patient's extremities were unduly cool. Only on a repeat study with sufficient warming were diffuse fibrillation potentials identified, helping to establish a diagnosis of motor neuron disease.

The effects of temperature variation on the configuration of a given positive wave or fibrillation potential remains uncertain. Extrapolating from the effects of temperature variation on a MUAP, one would anticipate that focal cooling of muscle would increase the duration and amplitude of a given positive wave or fibrillation potential.

Fasciculations. The effect of temperature variation on fasciculation potentials is likely complex, as the potentials can be generated both proximally and distally within a nerve.[22,130] As in other situations, the form of cooling applied may play a significant role in the temperature-associated changes. For example, if only the muscle were cooled, the frequency of a discharging motor unit might not change, but the duration of the fasciculation potential would be expected to increase, similar to that of a single motor unit. Indeed, Denys[31] noted that the changes in amplitude and shape of a fasciculation potential recorded from the surface are similar to those seen in the overall CMAP with focal cooling of muscle. On the other hand, if only the region of nerve from which the discharge was emanating is cooled, one might anticipate that the frequency of the discharge would decrease with no change in morphology. This is likely, as cooling generally tends to inhibit spontaneous depolarizations, similar to the decrease in the number of fibrillation potentials present with cooling.

Other Forms of Abnormal Spontaneous Activity. There has been little study to determine the effects of temperature variation on other neuronally generated spontaneous activity, including myokymic, neuromyotonic, and cramp discharges. One study suggested increased stiffness due to cooling in neuromyotonia; however, it was unclear from the data presented whether the patient actually had a myotonic disorder instead.[26] There are also two reports of patients who had worsened symptomatic cramps with heat exposure and exercise, but it is most likely that these patients had muscle contractures possibly due to metabolic myopathy rather than neuronally generated cramps.[7,109] With all neuronal discharges, it seems likely that cooling of the muscle will increase the amplitude and duration of the wave forms, akin to what occurs with fasciculations, whereas cooling of the nerve might inhibit spontaneous firing and reduce the discharge frequency. There are also no published data on the effect of temperature on complex repetitive discharges. However, direct cooling of muscle might be expected to inhibit their occurrence.

Demyelinating Disorders. *Conduction Block.* The effect of temperature modulation in demyelinating diseases is especially interesting. Increasing temperature in patients with central nervous system demyelination (e.g., multiple sclerosis) may increase symptoms, causing previously "silent" lesions to become manifest (Uhthoff's phenomenon).[87,127] Cooling, on the other hand, may reduce symptoms and improve strength, vision, or sensation in these patients. The mechanism for this temperature sensitivity was investigated in the early 1970s separately by Rasminsky,[107] examining rat ventral root single neurons treated with diphtheria toxin, and Davis and Jacobson, [24] studying sciatic nerve crush preparations in the guinea pig. The authors were able to demonstrate that, with heating, demyelinated neurons de-

veloped reversible conduction block at lower temperatures than normal neurons.

The mechanism underlying the development of temperature-sensitive conduction block again relies on changes in Na^+ channel open time. In areas of demyelination, there is reduced resistance and increased capacitance. Hence, current tends to "leak" out between nodes, leaving less current available to excite the downstream node. Normally, there is five times as much current presented to the node than is necessary to induce its depolarization. This is the so-called "axonal safety factor" (in contrast to the safety factor associated with neuromuscular transmission). In demyelinated neurons, this safety factor may decrease considerably. For example, demyelination could produce a situation in which there is barely sufficient current to depolarize the downstream node (i.e., a safety factor of about 1.0). Now, if the temperature of that demyelinated neuron were increased, Na^+ channels would open for a shorter time period, inducing a weaker current. If the current decreased sufficiently, such that the safety factor were below 1.0, then conduction block would ensue in that neuron.

The effect of temperature on peripheral nerve diseases with demyelinating components has obtained very little attention until relatively recently. Given that the studies by Rasminsky and Davis and Jacobson were actually performed on peripheral nerve preparations, it seems reasonable to expect that demyelinating peripheral nerve diseases could have similar temperature sensitivity. In fact, this does appear to be the case. Chaudry and coworkers[21] described a child with a form of chronic inflammatory demyelinating polyradiculoneuropathy who was gradually improving without treatment until she developed an upper respiratory tract infection associated with high fever. This produced prominent generalized weakness that improved with the use of antipyretics alone. Electrophysiologic studies of the patient compared with control subjects suggested the emergence of conduction block with heating. Likewise, Franssen and colleagues[49] described several patients with inflammatory demyelinating lesions that worsened with heating and improved with cooling. Additionally, Magy and colleagues[97] reported a family with a form of chronic inflammatory demyelinating polyneuropathy that appeared to be especially temperature sensitive, in which exacerbation of the disorder occurred with fever.

Compression neuropathies also have a demyelinating component. Indeed, Wang and colleagues[128] found that heat enhanced conduction block in sensory fibers in patients with carpal tunnel syndrome. Likewise, others[49,112] have found that heating of the arm can induce conduction block in motor fibers in patients with ulnar neuropathy at the elbow (Fig. 3).

Conduction Velocity. Theoretically, CV should increase with increasing temperature, regardless of the presence of demyelination. Indeed, Rasminsky[107] demonstrated that CV in demyelinated rat ventral root axons increased up until the moment that conduction block ensued. However, as noted above, thinly myelinated fibers have less temperature sensitivity than do thickly myelinated fibers, so a muted

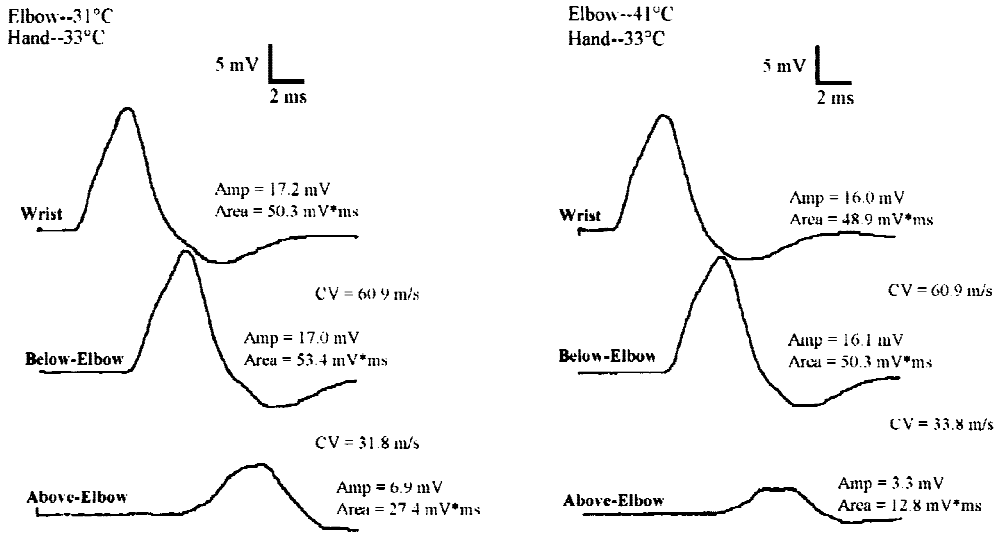


FIGURE 3. Heat-induced conduction block in a patient with ulnar neuropathy at the elbow. **(A)** Baseline study. **(B)** Study performed with the elbow submerged in a 45°C water bath for 20 min.

temperature effect in demyelinated neurons might be anticipated. In fact, this does appear to be the case. Davis and colleagues[25] demonstrated such an effect on guinea pig nerves with demyelination induced through experimental allergic neuritis. Dioszeghy and Stålberg[34] noted a decreased temperature sensitivity in several patients with diabetic demyelinating polyneuropathy, although in some other disorders with a demyelinative component the authors did not find a difference. Likewise, Notermans and colleagues[103] found a reduced temperature coefficient of 1.1 m/s per degree Celsius in patients with polyneuropathy associated with gammopathy. In Wang and colleagues' study of patients with carpal tunnel syndrome,[128] sensory CV increased only 0.7 m/s per degree Celsius on average. Ashworth and colleagues[2] found an even more reduced sensitivity in carpal tunnel syndrome patients, with sensory CV increasing only 0.1 m/s per degree Celsius.

Muscle Diseases. ***Nerve Conduction Studies.*** In patients with myopathic conditions without myotonia, temperature modulation is unlikely to have any significant effect on electrophysiologic parameters as compared with a normal individual, although this has not been studied specifically. In patients with myotonic conditions, the situation is more complex and is likely specific for a given disease. For example, with repetitive stimulation, patients with myotonic disorders often show a decline in CMAP amplitude, due to some of the muscle fibers becoming refractory with each stimulus.[124] If cooling enhances symptomatic myotonia, then repetitive stimulation will reduce the size of the CMAP even further. In patients with paramyotonia congenita, with sufficient cooling, the muscle becomes entirely inexcitable, hence reducing the CMAP amplitude to zero.

Electromyography.

Motor Unit Action Potentials. Effects are similar to that observed in normal muscle.

Fibrillation Potentials and Positive Sharp Waves. The same effects are likely to hold as described above.

Myotonic Discharges. A myotonic discharge, similar to a fibrillation or positive wave, represents the spontaneous, repetitive depolarization of a muscle fiber. Unlike a fibrillation potential or positive sharp wave, however, the amplitude and frequency of this wave form fluctuate, and the firing rate is often faster. Disorders associated with myotonia are associated with a decreased resting membrane potential. In myotonia congenita, this is due to a defective chloride channel,[79] whereas in paramyotonia congenita, this is secondary to a defective Na^+ channel, with an abnormally high permeability at rest.[39] In myotonic dystrophy and proximal myotonic myopathy, the defect remains uncertain, although dysfunction in a protein kinase may play a role.[16]

In some myotonic disorders, cooling increases the abundance of myotonic discharges on needle EMG. This likely occurs in myotonia congenita, as described by Nielsen and colleagues.[102] However, patients generally do not complain of cold-induced weakness. In paramyotonia congenita, there is also a strong temperature sensitivity, with cooling increasing the profusion of myotonic discharges until the muscle is cooled sufficiently that complete electrical silence occurs, usually at approximately 18°C.[102] This is due to a temperature-associated opening of mutant Na^+ channels.[88] Even though roughly half the Na^+ channels in a given muscle fiber are normal (as in any affected individual one of the two genes coding for this protein is normal) once the abnormal channels open, the membrane depolarizes sufficiently to induce opening of even the normal channels.

On the other hand, in some situations, myotonic discharge frequency will actually increase with warming. For example, myotonic activity induced by aromatic monocarboxylic acids appears to increase with heat.[4] Likewise, an increase in myotonia with heating has been reported in two patients with proximal myotonic myopathy.[115] Surprisingly, specific data on the effect of temperature on the most common myotonic disorder, myotonic dystrophy, are lacking.

Disorders of Neuromuscular Transmission. ***Myasthenia Gravis.*** The effect of temperature in patients with myasthenia gravis has been studied extensively. An exacerbation of symptoms with heat has long been appreciated.[55,120,121] Applying an ice pack to the eye improves ptosis,[43,118] whereas a hot pack worsens it.[80] In addition, authors have noted the importance of keeping an extremity warm while performing repetitive stimulation.[12] Rutkove and colleagues[114] also have demonstrated greatly enhanced decrements with repetitive stimulation at high temperature (Fig. 4). One proposed explanation for the predominantly proximal involvement of muscles in the disease is that severity of disease parallels the temperature of affected muscles, with extraocular muscles being the warmest.[73] Immunohistological and electromyographic abnormalities appear to follow a similar pattern.[122,126]

In terms of single-fiber EMG, there can be no doubt that with heating there will be increased blocking of impulses in patients with myasthenia

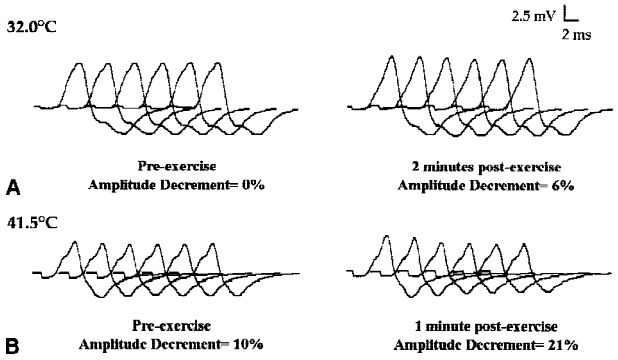


FIGURE 4. Repetitive stimulation (3 Hz) before and after exercise in a patient with myasthenia gravis. **(A)** Baseline study at 32°C. A decremental response is not observed either pre- exercise or postexercise. **(B)** Study at 41.5°C. A decrement becomes prominent even without exercise.

gravis. Although it would seem to follow logically that jitter should also increase with heating, as has been suggested,[122] this actually is unproven and would be opposite that which occurs in normal individuals.

The prominent effect of heat worsening the symptoms of myasthenia gravis and cold ameliorating them is due to many of the factors described above in the section on normal physiology. With cooling, for example, activity of acetylcholinesterase is reduced,[47,119] and the amplitude of individual ACh-induced depolarizations is increased (again, due to increased Na^+ influx).[60,85] Moreover, although cooling may reduce the rate of mobilization of secondary stores of ACh, it also increases facilitation secondary to a prolonged half-life of Ca^{++} in the nerve terminal, allowing more ACh to be released with subsequent stimuli.[28] Given the multiple reasons for an improvement in synaptic function, it is not surprising that lower temperature produces a reduced decrement on repetitive stimulation.

Lambert–Eaton Myasthenic Syndrome. Temperature sensitivity similar to that seen in myasthenia gravis occurs in Lambert–Eaton myasthenic syndrome. Cool temperatures will improve motor response amplitude at baseline and prolong the effect of posttetanic facilitation.[33,108,129] In one study, the effect of brief exercise in combination with 6–12°C cooling prolonged facilitation by about 25%.[96] These effects likely are due to enhanced release of ACh due to a prolonged half-life of Ca^{++} with cooling.[28] Based on these data, it would appear that, like myasthenia gravis, keeping an extremity warm will improve the diagnostic sensitivity of electrophysiologic testing.

Botulism. Once again, cooling appears to improve neuromuscular junction function in patients with botulism. Lundh[95] studied the effect of cooling in rats exposed to botulinum toxin. He demonstrated a remarkably strong inverse relationship between EPP amplitude and temperature in botulinum-exposed mice as compared with normal controls.

Table 2. Temperature effects in disease.

Disease type	Effect of temperature
Axonal loss lesions	
CMAP amplitude	Increased sensitivity to cooling
Abundance of fibrillation potentials	Decrease with muscle cooling
Fasciculation potential amplitude	Increase with muscle cooling
Fasciculation potential frequency	Likely decrease with neuronal cooling
MUAP amplitude	Insufficient data
Demyelinating lesions	
Conduction velocity	Reduced effect of temperature
Conduction block	Increases with heating
Neuromuscular junction disorders	
Myasthenia gravis	Improvement in decrement with cooling
Lambert–Eaton myasthenic syndrome	Baseline amplitude increases with cooling
Muscle disorders	
Nonmyotonic myopathy CMAP	Insufficient data
Paramyotonia congenita CMAP	Decreased amplitude with muscle cooling
Myotonic discharges in:	
Paramyotonia congenita	Increase, then abate with cooling
Myotonia congenita	Increase with cooling
Proximal myotonic myopathy	Increase with heating
Myotonic dystrophy	Insufficient data

Drugs. Studies have demonstrated that cooling will enhance the effect of drugs with a depolarizing action, such as methonium, whereas in those that competitively bind with ACh receptor, cooling will have the opposite effect, helping to maintain neuromuscular transmission.[19]

Organophosphates can produce a number of physiological abnormalities, the most prominent of which is a spontaneous repetitive motor action potential (SRMAP) that follows the CMAP.[10] The mechanism of this has been debated, but it is believed due to a repetitive action potential generated in the nerve terminal, possibly due to high levels of synaptic ACh interacting with presynaptic receptors. Rutchik and Rutkove[110] studied the effects of tem-

perature on a single patient with organophosphate intoxication from a suicide attempt. With heating, this secondary response was reduced in size relative to the CMAP. More importantly, cooling inhibited the SRMAP completely, most likely due to the increased refractory period of nerve and muscle at cold temperature.

NEURONAL DYSFUNCTION AND INJURY DUE TO TEMPERATURE EXTREMES

Thus far, the discussion has focused on mild to moderate changes in temperature. However, it is important to recognize that with more extreme heating and cooling, additional effects occur, producing dramatic changes in neuronal function, many of which may be irreversible.

Cold Injury. Cold injury to nerves, including trench foot and frostbite, has been appreciated since the time of Hippocrates and usually arises in conjunction with military activities, such as Napoleon's disastrous Russian Campaign, the First and Second World Wars, and, most recently, the Falklands War.[106] Cooling has also produced a few cases of iatrogenic focal mononeuropathy. For example, phrenic nerve injury has occurred as a consequence of cooling the heart for surgery.[20,41] Likewise, several lower-extremity mononeuropathies, including peroneal neuropathies, have occurred due to the overzealous application of ice for sports injuries.[5]

The mechanism of cold-induced nerve injury has been debated, but several early investigators raised the possibility of an ischemic origin.[30,51,84] Experimental proof for this was lacking until a recent study by Jia and Pollock[75] in which the investigators measured nerve blood flow in the sciatic nerve of rats using a laser Doppler flowmeter during cooling down to as low as 1°C for 3 h and then with rewarming. The authors documented dramatic reductions in blood flow within 50 min of the onset of cooling down to approximately 25% of normal. Even after rewarming at the end of 3 h, sciatic blood flow remained at only 39% of normal. In addition, utilizing a dissecting microscope, the authors identified slowing of blood flow through arterioles, metarterioles, capillaries, and venules. Vessel walls became irregular, and the diameter of the capillaries was reduced. The authors proposed a number of contributing factors that likely interplay to produce this phenomenon, including increased blood viscosity, constriction of arterioles and metarterioles, and capillary endothelial cell edema.

Reperfusion injury likely also plays a role. It has long been appreciated that fluctuations in temperature may predispose to frostbite injury. For example, warming one's cold feet by a fire followed by reexposure to cold may actually enhance the development of frostbite, as compared with continuous cold exposure. In additional work, Jia and Pollock[74] demonstrated that nerves exposed to intermittent cooling were more likely to suffer deleterious effects than were nerves exposed to continuous cold. Using the same model, the authors demonstrated that continuous cooling for 2 h produced more mild injury than did 1 h of cooling, followed by 1 h of rewarming, followed by another h of cooling. In the continuously cooled rats, compound action potentials of A-fibers recovered to some extent (about 20% of baseline) within 1–2 h of the end of cooling. In those exposed to intermittent cooling, no recovery of A-fiber function occurred, even after 6 days. Intramyelinic edema and swollen endothelial cells, present in the pathological examination of this study, are also consistent findings in nerves injured by reperfusion.[104] In other studies of rabbit hind-limb cold immersion, the site of severe injury has been confined to a small region of the upper tibial nerve at a level just below the water line, identified as a warm/cold interface by serial temperature measurements.[72,76]

In contrast to these findings, Hodgkin and Katz[66] showed that the giant axon of the squid continued to function down to −1°C, prompting the authors to comment, "It seems unlikely that a fresh squid nerve can be blocked above the freezing-point of sea water (−1.88°C)." The work was performed in vitro rather than in vivo, like the studies above, hence removing the issue of ischemic injury. Indeed, in all studies published since that time that have used in vivo techniques, conduction block has occurred with cooling down to the 3–10°C range.[18,50,92,105] In short, cooling induces ischemic block, but it has not been demonstrated to produce block by impeding ion channel function directly.

Contrary to the above studies, there are several reports that suggest that cooling may have a beneficial effect on peripheral nerves. For example, Sea and colleagues[117] found that the use of cooling could slow the rate of degeneration of transected myelinated axons in the ventral tail of the rat from 3 days at 32°C to 10 days at 13°C. In addition, Mitsui and colleagues[100] demonstrated that mild cooling (28°C) could be used to reduce the detrimental effects of ischemia in rat nerve. Rat nerves exposed to ischemia at cooler temperatures remained more functionally and structurally intact than did those exposed at 36°C. Although in seeming contradiction to the work described above, it is likely that the de-

gree of cooling (for example, mild in Mitsui's study) accounts for these differing results.

Heat Injury. Exposure of nerves to elevated temperature can also produce significant injury. This was first identified in studies of the giant axon of the squid performed by Hodgkin and Katz,[66] who noted that warming the axon to 35–40°C prevented impulse propagation. Studies have since demonstrated that warming mammalian nerves into the range of 42–48°C can produce reversible conduction block. In this situation, Na^+ channels open and close so rapidly that insufficient current is generated to excite the adjacent channels. Whether myelinated or unmyelinated fibers are more sensitive to this form of conduction block is unclear, as authors have found varied results.[78,89] In addition, one study also suggested that sensory fibers are more subject to heat-induced conduction block than are motor fibers.[42] In more recent work, Xu and Pollock[131] described the onset of reversible conduction block in rat sciatic unmyelinated neurons at approximately 47°C; at higher temperatures (e.g., 58°C), immediate and selective axonal degeneration of unmyelinated fibers occurred. In contrast, although heating up to 58°C did not produce immediate destruction of myelinated fibers, heating to relatively lower temperatures (47°C) resulted in a delayed selective failure of myelinated nerve fiber conduction after the temperature had returned to normal. This loss of predominantly myelinated fibers was due to a heat-induced angiopathy in the vasa nervorum, producing a marked reduction in blood flow with ischemia. The fact that myelinated fibers were injured at this low temperature is consistent with unmyelinated fibers being more ischemia resistant. Meanwhile, other studies have suggested a direct effect of moderate heat (44°C) on the myelin sheath.[69]

Nerve injuries, including compression neuropathies and generalized polyneuropathy, have also been reported in burn patients.[61,77,99] In one study of 800 burn victims over 7 years, 14 patients had single or multiple mononeuropathies confirmed on needle EMG testing; only one patient had a generalized polyneuropathy. Direct thermal injury and vascular occlusion were suggested possible etiologies.[99] On the other hand, some studies have found evidence of mild neuronal dysfunction in limbs without burn injury.[62,98] The mechanism for such effects is poorly understood and may be due to a number of factors, including critical illness, metabolic derangement, compression, or a generalized inflammatory reaction.[98]

REFERENCES

1. Abramson DI, Chu L, Jr, ST, Lee SW, Richardson G, Levin M. Effect of tissue temperatures and blood flow on motor nerve conduction velocity. JAMA 1966;198:1082–1088.
2. Ashworth NL, Marshall SC, Satkunam LE. The effect of temperature on nerve conduction parameters in carpal tunnel syndrome. Muscle Nerve 1998;21:1089–1091.
3. Balnave RJ, Gage PW. Temperature sensitivity of the time course of facilitation of transmitter release. Brain Res 1970; 21:297–300.
4. Barchi R. Muscle membrane chloride conductance and the myotonic syndromes. In: Cobb W, VanDuijn H, editors. Contemporary clinical neurophysiology. Amsterdam: Elsevier; 1978. pp 559–570.
5. Bassett FH, Kirkpatrick JS, Engelhardt DL, Malone TR. Cryotherapy-induced nerve injury. Am J Sports Med 1992;20: 516–518.
6. Behse F, Buchthal F. Normal sensory conduction in the nerves of the legs in man. J Neurol Neurosurg Psychiatry 1971;34:404–414.
7. Bergeron MF. Heat cramps during tennis: a case report. Int J Sport Nutr 1996;6:62–68.
8. Bergh U, Ekblom B. Influence of muscle temperature on the maximal muscle strength and power output in human skeletal muscles. Acta Physiol Scand 1979;107:33–37.
9. Bertram MF, Nishida T, Minieka MM, Janssen I, Levy CE. Effects of temperature on motor unit action potentials during isometric contraction. Muscle Nerve 1995;18:1443–1446.
10. Besser R, Gutmann L, Dillmann U, Weilmann LS, Hopt HC. End plate dysfunction in acute organophosphate intoxication. Neurology 1989;39:561–567.
11. Bolton CF, Sawa GM, Carter K. The effect of temperature on human compound action potentials. J Neurol Neurosurg Psychiatry 1981;44:407–413.
12. Borenstein S, Desmedt JE. New diagnostic procedures in myasthenia gravis. In: Desmedt J, editor. New developments in electromyography and clinical neurophysiology. Basel: Karger; 1973. p 350–374.
13. Boyd IA, Martin AR. The endplate potentials in mammalian muscle. J Physiol (Lond) 1956;132:74–91.
14. Brodie C, Sampson SR. Contribution of electrogenic sodium-potassium ATPase to resting membrane potential of cultured rat skeletal myotubes. Brain Res 1985;347:28–35.
15. Bromberg MB, Scott DM, the Ad Hoc Committee of the AAEM Single Fiber Special Interest Group. Single fiber EMG reference values: reformatted in tabular form. Muscle Nerve 1994;17:820–821.
16. Brook JD, McCurrach ME, Harley HG, Buckler AJ, Church D, Aburatani H, Hunter K, Stanton VP, Thirion JP, Hudson T, Sohn R, Zemelman B, Snell RG, Rundle SA, Crow S, Davies J, Shelbourne R, Buxton J, Jones C, Juvonen V, Johnson K, Harper PS, Shaw DJ, Houseman DE. Molecular basis of myotonic dystrophy: expansion of a trinucleotide (CTG) repeat at the 3′ end of a transcript encoding a protein kinase family member. Cell 1992;68:799–808.
17. Buchthal F, Pinelli P, Rosenfalk P. Action potential parameters in normal human muscle and their physiological determinants. Acta Physiol Scand 1954;32:219–229.
18. Byck R, Goldfard J, Schaumburg HH, Sharpless SK. Reversible differential block of saphenous nerve by cold. J Physiol (Lond) 1972;222:17–26.
19. Cannard TH, Zaimis E. The effect of lowered muscle temperature on the action of neuromuscular blocking drugs in man. J Physiol (Lond) 1959;149:112–117.
20. Chandler KW, Rozas CJ, Kory RC, Goldman AL. Bilateral diaphragmatic paralysis complicating local cardiac hypothermia during open heart surgery. Am J Med 1984;77:243–249.
21. Chaudry V, Crawford TO, DeRossett SE. Thermal sensitivity in demyelinating neuropathy. Muscle Nerve 1993;16: 301–306.

22. Conradi S, Grimby L, Lundemo G. Pathophysiology of fasciculations in ALS as studied by electromyography of single motor units. Muscle Nerve 1982;5:202–208.
23. Davies CTM, Mecrow IK, White MJ. Contractile properties of the human triceps surae with some observation on the effects of temperature and exercise. Eur J Appl Physiol 1982; 49:225–269.
24. Davis FA, Jacobson S. Altered thermal sensitivity in injured and demyelinated nerve: a possible model of temperature effects in multiple sclerosis. J Neurol Neurosurg Psychiatry 1971;34:551–561.
25. Davis FA, Schauf CL, Reed BJ, Kesler RL. Experimental studies of the effects of extrinsic factors on conduction in normal and demyelinated nerve. I. Temperature. J Neurol Neurosurg Psychiatry 1975;39:442–448.
26. de Carvalho M. Cold-induced neuromyotonia. J Neurol 1996;243:614–616.
27. De Jesus PV, Hausmanowa-Petrusewicz I, Barchi RL. The effect of cold on nerve conduction of slow and fast nerve fibers. Neurology 1973;23:1182–1189.
28. Delaney KR, Tank DW. A quantitative measurement of the dependence of short-term synaptic enhancement on presynaptic residual calcium. J Neurosci 1994;14:5885–5902.
29. Delbeke J, Kopec J, McComas AJ. Effects of age, temperature, and disease on the refractoriness of human nerve and muscle. J Neurol Neurosurg Psychiatry 1978;41:65–71.
30. Denny-Brown D, Adams RD, Brenner C, Doherty MM. The pathology of injury to nerve induced by cold. J Neuropathol Exp Neurol 1945;4:305–323.
31. Denys EH. AAEM Minimonograph #14: The influence of temperature in clinical neurophysiology. Muscle Nerve 1991;14:795–811.
32. Denys EH. The effect of temperature on the compound action potential in neuromuscular disease and normal controls. Electroencephalogr Clin Neurolphysiol 1977;43:598.
33. Denys EH. M wave changes with temperature in amyotrophic lateral sclerosis and disorders of neuromuscular transmission. Muscle Nerve 1990;13:613–617.
34. Dioszeghy P, Stålberg E. Changes in motor and sensory nerve conduction parameters with temperature in normal and diseased nerve. Electroencephalogr Clin Neurophysiol 1992;85:229–235.
35. Dlouha H, Donselaar Y, Teisinger J, Vyskocil F. Effect of temperature and oubain on the Na^+-K^+ activated membrane ATPase and electrogenic ionic pump of the golden hamster and mouse diaphragm. Physiol Bohemoslov 1980;29: 543–552.
36. Douglas WW, Malcolm JL. The effect of localized cooling on conduction in cat nerves. J Physiol (Lond) 1955;130:53–71.
37. Draper DO, Castel JC, Castel D. Rate of temperature increase in human muscle during 1 MHz and 3 MHz continuous ultrasound. J Orthop Sports Phys Ther 1995;22:142–150.
38. Dumitru D, DeLisa JA. Volume conduction. Muscle Nerve 1991;1991:605–624.
39. Ebers GC, George AL, Barchi RL, Ting-Passador SS, Kallen RG, Lathrop GM, Beckmann JS, Hahn AF, Brown WF, Campbell RD. Paramyotonia congenita and hyperkalemic periodic paralysis are linked to the adult muscle sodium channel gene. Ann Neurol 1991;30:810–816.
40. Edelwejn Z. Trials of evaluation of conduction velocity in motor fibers of different size under conditions of local hyperthermia. Acta Physiol Pol 1964;15:503–511.
41. Efthimiou J, Butler J, Woodham C, Benson MK, Westaby S. Diaphragm paralysis following cardiac surgery: role of phrenic nerve cold injury. Ann Thorac Surg 1991;52: 1005–1008.
42. Eliasson SG, Monafo WW, Nussbaum G, Olsen L. Differential effects of in vitro heating on rat sciatic nerve branches and spinal nerve roots. Exp Neurol 1986;93:57–66.
43. Ertas M, Arac N, Kumral K, Tuncbay T. Ice test as a simple diagnostic aid for myasthenia gravis. Acta Neurol Scand 1994;89:227–229.
44. Falck B, Lang H. Effects of temperature on motor unit potentials. Muscle Nerve 1986;9:573–574.
45. Fatt P, Katz B. Spontaneous subthreshold activity at motor nerve endings. J Physiol (Lond) 1952;117:109–128.
46. Feinstein B, Pattle RE, Weddell CT. Metabolic factors affecting fibrillations in denervated muscle. J Neurol Neurosurg Psychiatry 1945;8:1–16.
47. Foldes FF, Kuze S, Vizi ES, Deerv A. The influence of temperature on neuromuscular performance. J Neurol Trans 1978;43:27–45.
48. Franssen H, Wieneke GH. Nerve conduction and temperature: necessary warming time. Muscle Nerve 1994;17: 336–344.
49. Franssen H, Wieneke GH, Wokke JHJ. The influence of temperature on conduction block. Muscle Nerve 1999;22: 166–173.
50. Franz DN, Iggo A. Conduction failure in myelinated and nonmyelinated axons at low temperatures. J Physiol (Lond) 1968;199:319–345.
51. Friedman NB. The pathology of trench foot. Am J Pathol 1945;21:387–433.
52. Gasser HS, Grundfest H. Axon diameters in relation to the spike dimensions and the conduction velocity in mammalian A-fibers. Am J Physiol 1939;127:393–414.
53. Gasser M, Trojaborg W. Clinical and electrophysiological study of pattern of conduction times in distribution of sciatic nerve. J Neurol Neurosurg Psychiatry 1964;27:351–357.
54. Goldman L, Albus JS. Computation of impulse conduction in myelinated fibers: theoretical basis of the velocity-diameter relation. Biophys J 1968;8:596–607.
55. Gutmann L. Heat-induced myasthenic crisis. Arch Neurol 1980;37:671–672.
56. Halar EM, DeLisa JA, Brozovich FV. Nerve conduction velocity: relationship of skin, subcutaneous, and intramuscular temperatures. Arch Phys Med Rehabil 1980;61:199–203.
57. Halar EM, DeLisa JA, Brozovich FV. Peroneal motor conduction velocity: the importance of temperature correction. Arch Phys Med Rehabil 1981;62:439–443.
58. Halar EM, DeLisa JA, Soine TL. Nerve conduction studies in upper extremities: skin temperature corrections. Arch Phys Med Rehabil 1983;61:412–416.
59. Halle JS, Scoville CR, Greathouse DG. Ultrasound's effect on the conduction latency of the superficial radial nerve in man. Phys Ther 1981;61:345–350.
60. Harris JB, Leach GDH. The effect of temperature on endplate depolarisation of the cat diaphragm produced by suxamethonium and acetylcholine. J Pharm Pharmacol 1968;20: 194–198.
61. Helm PA, Pandian G, Heck E. Neuromuscular problems in the burn patient: cause and prevention. Arch Phys Med Rehabil 1985;66:451–453.
62. Henderson B, Koepke GH, Feller I. Peripheral polyneuropathy among patients with burns. Arch Phys Med Rehabil 1971;52:149–151.
63. Henrikson JD. Conduction velocity of motor nerves in normal subjects and patients with neuromuscular disorders. Minneapolis, Minnesota: University of Minnesota; 1956.
64. Hill QV. The influence of temperature on the tension developed in isometric twitch. Proc R Soc Lond (Biol) 1951; 138:349–354.
65. Hlavova A, Abramson DI, Rickert BL, Talso JF. Temperature effects on duration and amplitude of distal median nerve action potential. J Appl Physiol 1970;28:808–812.
66. Hodgkin AL, Katz B. The effect of temperature on the electrical activity of the giant axon of the squid. J Physiol (Lond) 1949;109:240–249.
67. Hofmann WW, Parsons RL, Feigen GA. Effect of temperature and drugs on mammalian motor nerve terminals. J Physiol (Lond) 1966;211:135–140.

68. Holewijn M, Heus R. Effects of temperature on electromyogram and muscle function. Eur J Appl Physiol 1992;65: 541–545.
69. Hoogeveen JF, Troost D, vanderKracht AH, Wondergem J, Haveman J, Gonzalez D. Ultrastructural changes in the rat sciatic nerve after local hyperthermia. Int J Hyperthermia 1993;9:723–730.
70. Hovav G, Parnas H, Parnas I. Neurotransmitter release: facilitation and three-dimensional diffusion of intracellular calcium. Bull Math Biol 1992;54:875–894.
71. Hubbard JI, Jones SF, Landau EM. The effect of temperature change upon transmitter release, facilitation, and posttetanic potentiation. J Physiol (Lond) 1977;216:591–609.
72. Irwin MS. Nature and mechanism of peripheral nerve damage in an experimental model of non-freezing cold injury. Ann R Coll Surg Engl 1996;78:372–379.
73. Jablecki C, Benton A. The frequency of muscle involvement in myasthenia gravis correlates with mean muscle temperature. Muscle Nerve 1982;5:491–492.
74. Jia J, Pollock M. Cold nerve injury is induced by intermittent cooling. Muscle Nerve 1999;22:1644–1652.
75. Jia J, Pollock M. The pathogenesis of non-freezing cold injury. Observations in the rat. Brain 1997;120:631–646.
76. Kennett RP, Gilliatt RW. Nerve conduction studies in experimental non-freezing cold injury: II. Generalized nerve cooling by limb immersion. Muscle Nerve 1991;14:960–967.
77. Khedr EM, Khedr T, el-Oteify MA, Hassan HA. Peripheral neuropathy in burn patients. Burns 1997;23:579–583.
78. Klumpp D, Zimmermann M. Irreverisble differential block of A- and C-fibres following local nerve heating in the cat. J Physiol (Lond) 1980;298:471–482.
79. Koch MC, Steinmeyer K, Lorenz C, Ricker K, Wolf F, Otto M, Zoll B, Lehmann-Horn F, Grzeschik KH, Jentsch TJ. The skeletal muscle chloride channel in dominant and recessive human myotonia. Science 1992;257:797–800.
80. Komiyama A, Hirayama K. Diagnostic usefulness of a hot test in patients with mild ocular myasthenia gravis [Japanese]. No To Shinkei 1987;39:733–737.
81. Kramer JF. Effect of therapeutic ultrasound intensity on subcutaneous tissue temperature and ulnar nerve conduction velocity. Am J Phys Med 1985;64:1–9.
82. Lang AH, Puusa A. Dual influence of temperature on compound nerve action potential. J Neurol Sci 1980;51:81–88.
83. Langley JN. Observations on denervated and on regenerating muscle. J Physiol 1917;51:377–395.
84. Large A, Heinbecker P. Nerve degneration following prolonged cooling of an extremity. Ann Surg 1944;120:742–749.
85. Lass Y, Fischbach GD. A discontinuous relationship between the acetylcholine-activated channel conductance and temperature. Nature 1976;263:150–151.
86. Lee HJ, Kwon HK. Semiquantification of fibrillation potentials with intramuscular temperature drop using an animal model. J Korean Med Sci 1997;12:550–552.
87. Lepore FE. Uhthoff's symptom in disorders of the anterior visual pathways. Neurology 1994;44:1036–1038.
88. Lerche H, Mitrovic N, Dubowitz V, Lehmann-Horn F. Paramyotonia congenita: the R1448P Na+ channel mutation in adult human skeletal muscle. Ann Neurol 1996;39:599–608.
89. Letcher FS, Goldring S. The effect of radiofrequency current and heat on peripheral nerve action potential in the cat. J Neurosurg 1968;29:42–47.
90. Li CL. Effect of cooling on neuromuscular transmission in the rat. Am J Physiol 1958;194:200–206.
91. Liley AW. An investigation of spontaneous activity at the neuromuscular junction of the rat. J Physiol (Lond) 1956; 132:650–666.
92. Low PA, McLeod JG. Refractory period, conduction of trains of impulses, and effect of temperature on conduction in chronic hypertrophic neuropathy: electrophysiological studies on the Tremblar mouse. J Neurol Neurosurg Psychiatry 1977;40:434–447.
93. Lowitzsch K, Hopf HC, Galland J. Changes of sensory conduction velocity and refractory periods with decreasing tissue temperature in man. J Neurol 1977;216:181–188.
94. Ludin HP, Beyeler F. Temperature dependence of normal sensory nerve action potentials. J Neurol 1977;216:173–180.
95. Lundh H. Antagonism of botulinum toxin paralysis by low temperature. Muscle Nerve 1983;6:56–60.
96. Maddison P, Newsom-Davis J, Mills KR. Decay of postexercise augmentation in the Lambert–Eaton myasthenic syndrome. Neurology 1998;50:1083–1087.
97. Magy L, Birouk N, Vallat JM, Gouider R, Maisonobe T, Bouche P, Lyon-Caen O, Fontaine B. Hereditary thermosensitive neuropathy. Neurology 1997;48:1684–1690.
98. Margherita AJ, Robinson LR, Heimbach DM, Fishfader VL, Schneider VA, Jones D. Burn-associated peripheral polyneuropathy. A search for causative factors. Am J Phys Med Rehabil 1995;74:28–32.
99. Marquez S, Turley JJ, Peters WJ. Neuropathy in burn patients. Brain 1993;116:471–483.
100. Mitsui Y, Schmelzer JD, Zollman PJ, Kihara M, Low PA. Hypothermic neuroprotection of peripheral nerve of rats from ischaemia-reperfusion injury. Brain 1999;122:161–169.
101. Nakanishi T, Norris FH. Effect of local temperature on the resting membrane potential in rat muscles. Electroencephalogr Clin Neurophysiol 1970;28:633–636.
102. Nielsen VK, Friis ML, Johnsen T. Electromyographic distinction between paramyotonia congenita and myotonia congenita: effect of cold. Neurology 1982;32:827–832.
103. Notermans NC, Franssen H, Wieneke GH, Wokke JHJ. Temperature dependence of nerve conduction and EMG in neuropathy associated with gammopathy. Muscle Nerve 1994;17: 516–522.
104. Nukada H, Anderson GM, McMorran PD. Reperfusion nerve injury: pathology due to reflow and prolonged ischemia. J Peripher Nerv Syst 1997;2:60–69.
105. Paintal AS. Block of conduction in mammalian myelinated nerve fibres by low temperatures. J Physiol (Lond) 1965;180: 1–19.
106. Payne R. Lessons of the Falklands. World Med 1984;17: 26–27.
107. Rasminsky M. The effect of temperature on conduction in demyelinated single nerve fibers. Arch Neurol 1973;28: 287–292.
108. Ricker K, Hertel G, Stodieck S. Influence of temperature on neuromuscular transmission in myasthenia gravis. J Neurol 1977;216:273–282.
109. Rowland LP, Penn AS. Heat-related muscle cramps. Arch Intern Med 1974;134:1133–1134.
110. Rutchik JS, Rutkove SB. Effect of temperature on motor responses in organophosphate intoxication. Muscle Nerve 1998;21:958–960.
111. Rutkove SB. Pseudofacilitation: a temperature-sensitive phenomenon. Muscle Nerve 2000;23:115–118.
112. Rutkove SB, Geffroy MA, Lichtenstein SH. Heat-sensitive conduction block in ulnar neuropathy at the elbow. Clin Neuro Physiol 2001;112:280–285.
113. Rutkove SB, Kothari MJ, Shefner JM. Nerve, muscle, and neuromuscular junction electrophysiology at high temperature. Muscle Nerve 1997;20:431–436.
114. Rutkove SB, Shefner JM, Wang AK, Ronthal M, Raynor EM. High-temperature repetitive nerve stimulation in myasthenia gravis. Muscle Nerve 1998;21:1414–1418.
115. Sander HW, Tavoulareas GP, Chokroverty S. Heat-sensitive myotonia in proximal myotonic myopathy. Neurology 1996; 47:956–962.
116. Sargeant AJ. Effect of temperature on leg extension force and short-term power output in humans. Eur J Appl Physiol 1987;56:693–698.

117. Sea T, Ballinger ML, Bittner GD. Cooling of peripheral myelinated axons retards Wallerian degeneration. Exp Neurol 1995;133:85–95.
118. Sethi KD, Rivner MH, Swift TR. Ice pack test for myasthenia gravis. Neurology 1987;37:1383–1385.
119. Shukuya R. On the kinetics of human blood cholinesterase. II. The temperature effect upon cholinesterase activity. J Biochem 1953;40:135–140.
120. Simpson JA. Myasthenia gravis, a new hypothesis. Scot Med J 1960;5:419–436.
121. Simpson JA. Weather and myasthenic fatigue. Lancet 1974; 2:458–459.
122. Stålberg E. Clinical electrophysiology of myasthenia gravis. J Neurol Neurosurg Psychiatry 1980;43:622–633.
123. Stegeman DF, De Weerd JPC. Modelling compound action potentials of peripheral nerves in situ. II. A study of the influence of temperature. Electroencephalogr Clin Neurophysiol 1982;54:516–529.
124. Streib EW. AAEE Minimonograph #27: differential diagnosis of myotonic syndromes. Muscle Nerve 1987;10:603–615.
125. Torbergsen T, Stålberg E. Effects of temperature on muscle twitch. Muscle Nerve 1986;9:574.
126. Tsujihata M, Hazama R, Ishii N, Ide Y, Mori M, Takamori M. Limb muscle endplate in ocular myasthenia gravis; quantitative ultrastructural study. Neurology 1979;29:654–661.
127. Uhthoff W. Untersuchungen über die bei der Multiplen Herdsklerose vorkommenden Augenstörungen. Arch Psychiat Nervenkr 1890;21:303–410.
128. Wang AK, Raynor EM, Blum AS, Rutkove SB. Heat sensitivity of sensory fibers in carpal tunnel syndrome. Muscle Nerve 1999;22:37–42.
129. Ward CD, Murray MF. The effect of temperature on neuromuscular transmission in the Eaton–Lambert syndrome. J Neurol Neurosurg Psychiatry 1979;42:247–249.
130. Wettstein A. The origin of fasciculations in motor neuron disease. Ann Neurol 1979;5:295–300.
131. Xu D, Pollock M. Experimental nerve thermal injury. Brain 1994;117:375–384.